

The Spectral Shortcut Theorem: A Gradient-Geometric Explanation for Failure Modes of Joint Spectral-Spatial Deep Learning

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Abstract

[TODO: Write abstract last, after Phase 7.]

1 Introduction

The integration of spectroscopic imaging with deep learning has produced a class of compositional models that has become the de facto standard across infrared tissue classification, hyperspectral remote sensing, Raman imaging, and mass spectrometry imaging. A canonical pipeline processes a hyperspectral input $X \in \mathbb{R}^{H \times W \times S}$ in two stages: a per-pixel spectral reduction $f_{\theta} : \mathbb{R}^S \rightarrow \mathbb{R}^K$ followed by a spatial model $g_{\phi} : \mathbb{R}^{K \times H \times W} \rightarrow \mathbb{R}^{N_{\text{cls}} \times H \times W}$. End-to-end joint training of this pipeline is overwhelmingly the default — adopted by foundational work in FTIR histology Berisha et al. [2019], recent state-of-the-art spatial-spectral ViTs for prostate tissue classification O’Leary et al. [2026], and essentially all hyperspectral remote sensing systems.

An empirical paradox. Despite the prevalence of this paradigm, an unexpected and persistent pattern has emerged across multiple subfields. Frozen pretrained spectral encoders dramatically outperform end-to-end joint training on held-out test data. In our motivating tissue classification setting, a learned-from-scratch linear f_{θ} trained end-to-end achieves 0.675 test F1; freezing a pretrained f_{θ} yields 0.84–0.95 test F1. Most strikingly, replacing f_{θ} with *frozen random weights* yields 0.70 test F1 — outperforming the joint trained model. Recent independent observations O’Leary et al. [2026], Berisha et al. [2019] have noted, in different terms, that aggressive spectral compression to 16 features has negligible impact on spatial-spectral neural network performance, and have concluded that “tissue classification only needs a small set of spectral features.”

We argue the opposite. The reason end-to-end joint training fails is not that spectral information is unimportant. It is that *end-to-end optimization fails to extract that information*, even when the spectral signal is rich and discriminative. The problem is in the training procedure, not the data.

This paper. We provide a rigorous explanation for the failure of joint spectral-spatial training, grounded in the geometry of the loss landscape and the dynamics of stochastic gradient descent on compositional architectures with asymmetric capacity. Two main theorems form our contribution.

- **Theorem 11 (Hessian Capacity Bound).** We bound the condition number of the joint Gauss-Newton matrix in terms of the ratio of module capacities. Under asymmetry

$C_g \gg C_f$, the loss landscape is geometrically ill-conditioned with $\kappa(G) = \Omega(C_g/C_f)$. The proof combines block matrix analysis (Schur complement) with the mean-field FIM scaling of Karakida et al. [2019].

- **Theorem 26 (Two-Timescale Spurious Convergence).** The geometric ill-conditioning induces an implicit slow-fast dynamical system. The high-capacity spatial module rapidly fits the data using shortcuts in the (initially random) feature map, driving the cross-entropy residual to zero exponentially fast. The slow spectral module is gradient-starved before it can specialize. The system converges to a stationary point at which the spectral parameters have not moved appreciably from initialization. The proof combines Borkar’s two-timescale stochastic approximation Borkar [2008] with Pezeshki et al.’s gradient starvation framework Pezeshki et al. [2021].

A real-time diagnostic. The pathology described by Theorems 11 and 26 is invisible to standard loss curves — training and validation loss both decay smoothly while the spectral module is silently starved. To make the pathology observable in practice we introduce the *Effective Gradient Ratio* (EGR), a scalar diagnostic computable inside any training loop:

$$\text{EGR}(t) = \frac{\|\nabla_{\theta} \mathcal{L}(\theta_t, \phi_t)\|}{\|\nabla_{\phi} \mathcal{L}(\theta_t, \phi_t)\|}.$$

Proposition 29 establishes that EGR decays along the slow-fast manifold predicted by Theorem 26. Empirically, the depth of the EGR collapse scales monotonically with capacity ratio, providing a quantitative, training-time observable for gradient starvation.

Prescription. The theory yields a direct prescription: freeze the spectral module, or inject a fixed residual baseline. Freezing eliminates the timescale gap by removing θ from the optimization entirely. A residual baseline ($Z = f_{\theta}(X) + h(X)$ for a fixed h , e.g. PCA) is a softer intervention that breaks the shortcut without precluding spectral fine-tuning. We discuss both in Section 9.

Universality. Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We verify in Section 7 that the same pattern manifests across linear, CNN, and Vision Transformer spatial models on synthetic hyperspectral problems, indicating the phenomenon is architecture-agnostic. We expect the same mechanism to govern frozen-encoder practices in Raman spectroscopy, mass spectrometry imaging, NIR analysis, and beyond.

Relation to prior work. The closest prior work is in multimodal learning, where Wang et al. [2020] and Peng et al. [2022] identified “modality competition” and proposed gradient blending and modulation interventions. Our setting is unimodal (a single hyperspectral input) but the mechanism is structurally analogous: the spatial and spectral subspaces of the input act as competing “modalities,” with the spatial module monopolizing the gradient budget. We discuss this connection further in Section 2.

Outline. Section 2 reviews related work. Section 3 establishes notation. Section 4 proves Theorem 11. Section 5 proves Theorem 26. Section 6 formalizes the EGR diagnostic. Section 7 validates both theorems and the diagnostic on synthetic spectral-spatial problems with linear, CNN, and ViT spatial models. Section 8 briefly connects the synthetic findings to real FTIR/QCL tissue classification data. Section 9 discusses prescriptions, limitations, and open problems.

2 Related Work

Our analysis sits at the intersection of four lines of research: the implicit bias and gradient dynamics of cross-entropy training; the geometry of deep-network loss surfaces; multimodal gradient competition; and spectral-spatial deep learning for biomedical and remote-sensing applications. We survey each in turn.

2.1 Gradient starvation and implicit bias

Cross-entropy minimization is known to induce structural biases in the learned representation. Soudry et al. [2018] proved that on linearly separable data, gradient descent on logistic loss converges to the maximum-margin classifier. Gunasekar et al. [2017] and Lyu and Li [2020] extended this picture to deep homogeneous networks. These results establish that gradient-based training selects *which* solution among many that fit the training data.

Pezeshki et al. [2021] identified a distinct phenomenon termed *gradient starvation*: in the Neural Tangent Kernel regime, dominant features in the input rapidly saturate the cross-entropy loss, after which weaker but task-relevant features receive negligible gradient signal. Our Theorem 26 adapts this framework to compositional architectures, where the “dominant features” are introduced not by the input distribution but by an architectural capacity asymmetry between modules. The result is a module-level starvation rather than a feature-level one.

2.2 Two-timescale stochastic approximation

The classical machinery for analyzing coupled gradient flows with disparate timescales is Borkar’s stochastic approximation framework [2008]. Heusel et al. [2017] adapted this to neural network training by introducing the Two Time-Scale Update Rule for GANs, proving convergence to a local Nash equilibrium under explicit step-size disparity. Our setting differs structurally: the timescale separation in our compositional model is not imposed by the optimizer (the same learning rate is applied to both modules) but emerges *implicitly* from the Hessian eigenvalue gap quantified by Theorem 11.

2.3 Hessian geometry of deep networks

The eigenspectrum of the neural-network Hessian has been characterized empirically and theoretically. Sagun et al. [2017] identified a bulk-plus-outliers structure for overparametrized networks. Pennington and Bahri [2017] developed a random matrix theory analysis that predicts spectral densities under Gaussian assumptions. Karakida et al. [2019] derived closed-form mean-field expressions for the leading eigenvalue of the Fisher Information Matrix, showing it scales with the layer width and parameter count. Theorem 11 applies these results blockwise to the joint Hessian of a compositional model, combining them with Schur complement bounds to yield a capacity-ratio scaling for the joint condition number.

2.4 Multimodal gradient competition

The phenomenon most structurally analogous to ours is gradient competition in multimodal learning. Wang, Tran, and Feiszli [2020] showed empirically that joint training of multimodal classification networks underperforms unimodal training because different modalities overfit and generalize at different rates. They proposed Gradient Blending to dynamically reweight modality-specific losses. Peng et al. [2022] identified “modality laziness” — a tendency for one modality to dominate training, leaving the other under-optimized — and

proposed On-the-fly Gradient Modulation. These works are empirical; our contribution adapts and extends their framework to the unimodal *compositional* setting and provides a rigorous geometric foundation. The recent biomedical application of gradient blending to multimodal sarcoma prognosis Bozzo et al. [2024] confirms the mechanism operates in medical settings, but is restricted to explicit dual-modality (MRI + clinical) architectures, not the single-modality compositional pipelines we consider.

2.5 Spectral-spatial deep learning

Spectral-spatial deep learning has been an active area for over a decade, primarily in two communities: biomedical infrared spectroscopy and hyperspectral remote sensing. Berisha et al. Berisha et al. [2019] established the benefit of combining spectral and spatial features for FTIR histology and observed that some tissue types could be classified from spatial features alone, with “minimal chemical information” required — a foreshadowing of the spatial shortcut phenomenon we diagnose. More recently, O’Leary et al. O’Leary et al. [2026] systematically compared spatial-spectral architectures for prostate cancer classification in infrared spectroscopy and reported the striking finding that compressing the spectral dimension to as few as 16 features had negligible effect on all models tested. The authors interpreted this as evidence that tissue classification requires only a limited spectral signature. Our Theorem 26 provides an alternative explanation: end-to-end training fails to extract the spectral signature even when it is present.

The conceptually nearest work in biomedical imaging is the recent characterization of shortcut learning in histopathology by Dawood and Minhas et al. Dawood et al. [2026], who showed that AI models predicting molecular biomarkers from H&E images exploit spurious correlations with clinicopathological features rather than learning genuine biological signal. Our paper extends the shortcut-learning framework into a new modality (spectroscopic imaging) with a different mechanism (gradient competition driven by architectural asymmetry rather than confounder leakage).

2.6 Frozen and random features

Frozen random features have a long history in machine learning. Saxe et al. Saxe et al. [2011] showed that convolutional networks with random (untrained) filters achieve competitive performance on standard benchmarks. Rahimi and Recht Rahimi and Recht [2007] established the theoretical foundations of random features for kernel methods. More recently, Coil and Cheney Coil and Cheney [2025] showed that freezing layers acts as implicit regularization in overparametrized networks. These works establish that frozen random features are not an oddity but a principled choice. Our contribution is to identify the specific mechanism — gradient starvation under capacity asymmetry — under which freezing is not merely competitive but *strictly preferable* to joint training.

3 Setup and Preliminaries

We analyze a generic two-module compositional model. While our motivating application is infrared spectroscopic imaging for tissue classification, the framework applies to any pipeline that first reduces a per-element spectral signal and then applies a high-capacity spatial model. We deliberately keep the notation domain-agnostic.

3.1 The compositional model

Let $X \in \mathbb{R}^{H \times W \times S}$ denote a hyperspectral input image, where H and W are spatial dimensions and S is the spectral dimension.¹ We write the per-pixel spectrum at location (h, w) as $x_{h,w} \in \mathbb{R}^S$.

The model consists of two modules applied sequentially:

1. A *spectral reduction* $f_{\theta} : \mathbb{R}^S \rightarrow \mathbb{R}^K$ applied independently to every pixel, producing a feature map $Z \in \mathbb{R}^{K \times H \times W}$ with $Z_{:,h,w} = f_{\theta}(x_{h,w})$.
2. A *spatial model* $g_{\phi} : \mathbb{R}^{K \times H \times W} \rightarrow \mathbb{R}^{N_{\text{cls}} \times H \times W}$ producing per-pixel class logits $\hat{y} = g_{\phi}(Z)$.

We denote the composed model $F_{\theta,\phi}(X) = g_{\phi}(f_{\theta}(X))$ and the parameter counts $C_f := \dim(\theta)$ and $C_g := \dim(\phi)$.

Assumption 1 (Linear spectral reduction). f_{θ} is linear in θ : there exists $W \in \mathbb{R}^{S \times K}$ with $\theta = \text{vec}(W)$ and $f_{\theta}(x_{h,w}) = W^{\top} x_{h,w}$.

Remark 2. Assumption 1 captures the canonical case used widely in spectroscopy pipelines, including the FTIR and QCL tissue classification work we draw motivation from. It is also the cleanest case in which the Hessian of $\mathcal{L}$ admits closed-form block analysis. We discuss extensions to shallow nonlinear f_{θ} in Section 9 and verify empirically in Section 7 that the theorem’s scaling persists when f_{θ} is replaced by a small MLP.

Assumption 3 (Spatial model regularity). g_{ϕ} is twice continuously differentiable in ϕ with L -Lipschitz gradients.

Assumption 4 (Sub-Gaussian inputs). Per-pixel inputs $\{x_{h,w}\}$ are sub-Gaussian with second moment $\Sigma = \mathbb{E}[x_{h,w} x_{h,w}^{\top}]$ of bounded condition number.

Assumption 5 (Capacity gap). $C_g/C_f \rightarrow \infty$.

Assumption 5 is the structural condition under which our theorems become informative. Empirically it is satisfied with several orders of magnitude to spare in modern spectral-spatial architectures (in our motivating example, $C_g/C_f \approx 325$).

Assumption 6 (Bounded input Jacobian of the classifier). There exists a constant $\tilde{L} > 0$ depending on the depth of g_{ϕ} , the bottleneck dimension K , the number of classes N_{cls} , and the data distribution — but *not* on the characteristic width M or the total spatial parameter count C_g — such that

$$\sup_{\phi \in \Phi_{\text{reg}}, Z \in \mathcal{Z}_{\text{reg}}} \left\| \frac{\partial g_{\phi}(Z)}{\partial Z} \right\|_{\text{op}} \leq \tilde{L},$$

uniformly over the training-time parameter regime Φ_{reg} and the induced feature domain $\mathcal{Z}_{\text{reg}}$.

Remark 7 (Why Assumption 6 is separate from Assumption 3). Assumption 3 bounds the second derivatives of g_{ϕ} with respect to *parameters* ϕ (the standard L -smoothness condition of the optimization literature). Assumption 6 bounds the first derivative with respect to *input features* Z —a different object with a different constant. Parameter-side smoothness does not imply input-side Lipschitzness; we require both.

At random Kaiming/Xavier initialization, Marchenko–Pastur theory Marčenko and Pastur [1967] gives $\|\partial g_{\phi}/\partial Z\|_{\text{op}} = \Theta(1)$ in width M for standard CNN and MLP blocks. Under the maximal-update parameterization (μ P) of Yang and Hu Yang and Hu [2021], the bound persists throughout training as $M \rightarrow \infty$. Under standard-parameterization training with weight decay

¹Multi-channel inputs (e.g. raw absorbance together with first and second derivatives) are handled by concatenating channels along the spectral dimension; this changes only the input size and leaves the analysis unaffected.

$\lambda > 0$ and bounded-norm iterates, Assumption 6 follows from the boundedness of $\|\phi\|$ combined with the per-layer product-of-operator-norms bound; the constant $\tilde{L}$ depends multiplicatively on the depth L but not on the width. Softmax self-attention is Lipschitz only on bounded inputs Kim et al. [2021], so $\mathcal{Z}_{\text{reg}}$ should be taken compact whenever g_ϕ uses attention—this is guaranteed by LayerNorm before attention, which is standard practice.

3.2 Loss and gradients

We train with per-pixel cross-entropy:

$$\mathcal{L}(\theta, \phi) = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \text{CE}(F_{\theta, \phi}(X)_{:,h,w}, y_{h,w}), \quad (1)$$

where $y_{h,w} \in \{1, \dots, N_{\text{cls}}\}$ is the ground-truth class. Cross-entropy plays an essential role in Section 5: its gradient with respect to the logits $\hat{y}$ is the prediction residual $r = \text{softmax}(\hat{y}) - y_{\text{onehot}}$, which vanishes exponentially as predictions saturate. We discuss other loss functions sharing this vanishing-residual property in Section 9.

The gradient of $\mathcal{L}$ with respect to the spatial parameters is computed by standard back-propagation. The gradient with respect to the spectral parameters θ admits the chain-rule decomposition

$$\nabla_{\theta} \mathcal{L} = \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{y}}}_{\text{residual}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial Z}}_{\text{spatial Jacobian}} \cdot \underbrace{\frac{\partial Z}{\partial \theta}}_{\text{input}}. \quad (2)$$

The three factors play distinct roles in our analysis. The residual is small when predictions are confident. The spatial Jacobian depends on the current state of ϕ and thus changes throughout training. The final factor is the input itself and is stationary. The product structure implies that if the residual decays to zero, the gradient to θ vanishes regardless of the other two factors. This observation underlies Theorem 26.

3.3 The block Hessian

The joint Hessian of $\mathcal{L}$ decomposes into blocks aligned with the two parameter groups:

$$H = \nabla^2 \mathcal{L}(\theta, \phi) = \begin{pmatrix} H_{\theta\theta} & H_{\theta\phi} \\ H_{\phi\theta} & H_{\phi\phi} \end{pmatrix}, \quad (3)$$

with $H_{\theta\theta} \in \mathbb{R}^{C_f \times C_f}$, $H_{\phi\phi} \in \mathbb{R}^{C_g \times C_g}$, and cross blocks of appropriate sizes. Theorem 11 bounds the condition number of H in terms of C_g/C_f by separately bounding the spectrum of each block.

For optimization-rate analysis it is often more tractable to work with the Gauss-Newton approximation $G = J^\top J$, where $J = [J_\theta, J_\phi]$ is the Jacobian of the per-pixel residuals. The two matrices coincide near interpolation, and Theorem 11 is stated for G .

3.4 The Effective Gradient Ratio

We introduce a real-time diagnostic that tracks the asymmetry in gradient flow between the two modules.

Definition 8 (Effective Gradient Ratio). At training step t with parameters (θ_t, ϕ_t) , the Effective Gradient Ratio is

$$\text{EGR}(t) := \frac{\|\nabla_{\theta} \mathcal{L}(\theta_t, \phi_t)\|}{\|\nabla_{\phi} \mathcal{L}(\theta_t, \phi_t)\|}.$$

Remark 9. EGR is a unitless quantity directly computable inside any training loop with negligible overhead. Section 6 establishes its monotonicity along the slow-fast manifold (a consequence of Theorem 26) and proposes a threshold criterion for detecting gradient starvation onset.

3.5 Notational conventions

Throughout, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm for vectors and the spectral norm for matrices; $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the largest and smallest eigenvalues of a symmetric matrix; $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ its condition number. We write $A \succeq B$ to mean $A - B$ is positive semidefinite. The symbol $\Omega(\cdot)$ is used in the standard asymptotic sense.

Remark 10 (Effective condition number for rank-deficient Gauss-Newton matrices). The joint Gauss-Newton matrix G considered in Theorem 11 is typically rank-deficient in the bottleneck regime $N_{\text{cls}} < K$: parameter directions orthogonal to the classifier’s range have exactly zero curvature and receive no gradient. Consequently the strict smallest eigenvalue $\lambda_{\min}(G) = 0$ and the strict condition number is infinite. Throughout this paper we work instead with the *effective* condition number

$$\kappa_{\text{eff}}(G) := \frac{\lambda_{\max}(G)}{\lambda_{\min}^+(G)},$$

where $\lambda_{\min}^+(G)$ denotes the smallest *nonzero* eigenvalue—equivalently, the smallest eigenvalue of G restricted to its range. This is the ratio the optimizer actually experiences on the loss-sensitive subspace, and is standard in numerical linear algebra (Horn and Johnson Horn and Johnson [2013], §5.1) and in modern Hessian / NTK analyses of over-parameterised networks (Sagun et al. Sagun et al. [2017]; Ghorbani et al. Ghorbani et al. [2019]). All occurrences of $\kappa(G)$ in Theorem 11 and its proof are to be read as $\kappa_{\text{eff}}(G)$.

4 Theorem 1: The Hessian Capacity Bound

This section establishes our first main result: the condition number of the joint Gauss-Newton matrix grows with the ratio of module capacities. The result is the geometric foundation for the dynamical analysis of Section 5: an ill-conditioned loss landscape induces a forced timescale separation between the modules, which Theorem 26 then shows leads to gradient starvation.

4.1 Main result

Theorem 11 (Hessian Capacity Bound). *Let $G = J^\top J$ denote the Gauss-Newton matrix of the loss (1) at a parameter (θ, ϕ) . Under Assumptions 1–5, there exist constants $c_1, c_2 > 0$ depending only on the data covariance, the bottleneck dimension K , and the model class of g_ϕ such that*

$$\kappa_{\text{eff}}(G) \geq c_1 \cdot \sqrt{\frac{C_g}{C_f}} - c_2.$$

Equivalently, if g_ϕ has fixed depth and characteristic width M (so $C_g = \Theta(M^2)$ for typical MLP, CNN, ViT architectures),

$$\kappa_{\text{eff}}(G) \geq \tilde{c}_1 \cdot \frac{M}{\sqrt{C_f}} - c_2.$$

The proof decomposes the condition number bound into separate bounds on the largest and smallest eigenvalues of G . We state the three supporting lemmas before presenting the proof.

4.2 Supporting lemmas

Lemma 12 (Spatial block scaling). *Under Assumptions 3–4, let M denote the characteristic width of g_ϕ (channel count for a CNN, embedding dimension for a ViT, hidden layer width for an MLP). Then the largest eigenvalue of the spatial Gauss-Newton block satisfies*

$$\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(M) \quad \text{in expectation over random initialization.}$$

For fixed-depth g_ϕ , $C_g = \Theta(M^2)$, so equivalently

$$\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(\sqrt{C_g}).$$

Remark 13. Lemma 12 is a direct application of the mean-field analysis of Karakida et al. Karakida et al. [2019], who derive closed-form expressions for the leading eigenvalue of the Fisher Information Matrix of fully connected and convolutional networks with random Gaussian initialization. The scaling is linear in the parameter count when widths and depth are matched. We verify empirically in Section 7 that the scaling persists for the architectures we consider (linear, CNN, ViT).

Lemma 14 (Eigenvalue interlacing for block matrices). *For any symmetric block matrix $M = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$ with principal blocks A and D ,*

$$\lambda_{\max}(M) \geq \max\{\lambda_{\max}(A), \lambda_{\max}(D)\}.$$

Remark 15. Lemma 14 is a standard consequence of Cauchy’s interlacing theorem applied to principal submatrices (see e.g. Bhatia Bhatia [1997], Corollary III.1.5).

Lemma 16 (Residual-dimension rank bound and smallest-eigenvalue cap). *Let $J_\theta \in \mathbb{R}^{N_{\text{data}}HW N_{\text{cls}} \times C_f}$ denote the Jacobian of the softmax residual $r = \text{softmax}(\hat{y}) - y_{\text{onehot}}$ with respect to the spectral parameters θ , aggregated over N_{data} training examples and all HW pixels. Under Assumptions 1–6:*

(i) Rank bound. *The Gauss-Newton block of the spectral module has rank bounded by*

$$\text{rank}(J_\theta^\top J_\theta) \leq \min(C_f, N_{\text{data}} \cdot HW \cdot (N_{\text{cls}} - 1)).$$

In the over-parameterised regime $C_f > N_{\text{data}}HW(N_{\text{cls}} - 1)$, the block is rank-deficient with at least $C_f - N_{\text{data}}HW(N_{\text{cls}} - 1)$ eigenvalues exactly equal to zero.

(ii) Smallest positive eigenvalue cap. *There exists a constant $C_2 = C_2(\tilde{L}, \lambda_{\max}(\Sigma_X), N_{\text{cls}})$ independent of C_g (but depending on the depth of g_ϕ through $\tilde{L}$) such that*

$$\lambda_{\min}^+(J_\theta^\top J_\theta) \leq \frac{C_2}{N_{\text{data}}HW(N_{\text{cls}} - 1)}.$$

Remark 17. Lemma 16(i) is the per-pixel bottleneck extension of the classical Fisher-rank bound for softmax classification, going back to Amari Amari [2016] (rank- $(N_{\text{cls}} - 1)$ Fisher of the categorical family); the sample-level $N(N_{\text{cls}} - 1)$ formulation appears in Botev, Ritter, and Barber Botev et al. [2017], and the empirical spectrum observation is due to Sagun et al. Sagun et al. [2017] and Pappas Pappas [2020]. Our contribution is the extension to compositional networks with a linear per-pixel bottleneck and arbitrary spatial classifier, plus the explicit C_g -independence statement in part (ii).

Remark 18 (Gauss-Newton vs. full Hessian). Lemma 16 bounds the Gauss-Newton block $J_\theta^\top J_\theta$ (equivalently, the empirical Fisher). The full Hessian carries an additional residual-weighted second-derivative term $\sum_i r_i \cdot \partial^2 \ell_i / \partial \theta^2$ that vanishes at convergence (Sagun et al. Sagun et al. [2017]). The bound transfers to the full Hessian at any stationary point.

Remark 19. The Schur complement bound is classical; see Horn and Johnson Horn and Johnson [2013], Theorem 1.20. The second statement uses the linearity of f_θ to derive the Kronecker factorisation $J_\theta^\top J_\theta = A^\top A \otimes \Sigma_X$ (under a common-linearisation assumption on $A = \partial \hat{y} / \partial Z$); combined with the Bhatia spectral identity for Kronecker products, this yields the explicit bound $\lambda_{\min}^+(J_\theta^\top J_\theta) \leq L^2 \cdot \lambda_{\min}(\Sigma_X)$ where L is the classifier’s gradient Lipschitz constant. See Section A.1 for the full derivation.

4.3 Proof of Theorem 11

Proof of Theorem 11. The Gauss-Newton matrix admits the block decomposition

$$G = \begin{pmatrix} J_{\theta}^{\top} J_{\theta} & J_{\theta}^{\top} J_{\phi} \\ J_{\phi}^{\top} J_{\theta} & J_{\phi}^{\top} J_{\phi} \end{pmatrix}.$$

By Lemmas 12 and 14,

$$\lambda_{\max}(G) \geq \lambda_{\max}(J_{\phi}^{\top} J_{\phi}) = \Omega(\sqrt{C_g}),$$

using $C_g = \Theta(M^2)$ for fixed-depth architectures. By Lemma 16 (see Section A.1 for the full derivation), the smallest nonzero eigenvalue of the spectral Gauss-Newton block is capped by a C_g -independent constant C_2 depending only on the data covariance and the classifier’s Lipschitz constant. Transfer to the full matrix via a range-space argument (Section A.1) gives

$$\lambda_{\min}^+(G) \leq C_2.$$

Combining with the $\lambda_{\max}$ bound and identifying $C_f = SK$ (Assumption 1),

$$\kappa_{\text{eff}}(G) = \frac{\lambda_{\max}(G)}{\lambda_{\min}^+(G)} \geq \frac{c'_1 \cdot \sqrt{C_g}}{C_2} = c_1 \cdot \sqrt{\frac{C_g}{C_f}}$$

with $c_1 := c'_1 \cdot \sqrt{C_f}/C_2$ depending only on the data covariance, the bottleneck dimension K , the input spectral dimension S , and the model class of g_{ϕ} ; a constant $c_2 \geq 0$ absorbs finite- C_g corrections from Karakida’s concentration bound. This completes the proof. $\square$

4.4 Implications

Theorem 11 establishes that the loss landscape is *geometrically pathological* whenever $C_g \gg C_f$. Two immediate consequences follow.

Bounded learning rate forces timescale separation. The maximum stable learning rate for first-order optimization is bounded above by $2/\lambda_{\max}(G)$. Under Theorem 11, this bound scales as $\Omega(1/\sqrt{C_g})$. With such a small learning rate, updates to θ —whose Hessian block has smallest nonzero eigenvalue bounded above by a C_g -independent constant—progress at a rate that is $\Omega(\sqrt{C_g}/C_f)$ -fold slower than updates to ϕ . This is the dynamical regime analyzed in Section 5.

No remedy from optimizer tuning. The condition number bound depends only on architectural parameters, not on the optimizer or schedule. Adaptive methods (Adam, RMSProp) rescale gradients but cannot remove the underlying curvature asymmetry; they redistribute the timescale gap rather than eliminating it. We discuss this further in Section 9.

4.5 Comments on tightness

The bound in Theorem 11 is a *lower bound* on the condition number. In practice the empirical condition number can be substantially larger because $\lambda_{\min}(G)$ is often dominated by near-degenerate directions in the spectral block that the bound does not capture. Experiment 1.1 in Section 7 measures $\kappa(G)$ empirically across a range of capacity ratios and confirms that the predicted scaling holds with a slope close to the theoretical prediction.

Remark 20 (Defense against linear triviality). A natural objection is that, since f_{θ} is linear, the analysis must reduce to principal component analysis (PCA) of the input. This is not the case. PCA finds directions of maximum input variance *without reference to labels*. Theorem 11 concerns the curvature of the supervised cross-entropy loss, which depends on the current state of g_{ϕ} through the Jacobian $\partial \hat{y}/\partial Z$. The pathology we identify is precisely that this supervised signal fails to reach θ , not that f_{θ} is incapable of representing useful features.

5 Theorem 2: Two-Timescale Spurious Convergence

Theorem 11 establishes that the loss landscape is geometrically ill-conditioned under capacity asymmetry. This section translates that geometric pathology into a dynamical one: under joint SGD, the spectral parameters θ are gradient-starved by the spatial parameters ϕ and the system converges to a stationary point at which the spectral module has not specialized to any meaningful feature representation. The proof combines two-timescale stochastic approximation (Borkar Borkar [2008]) with the cross-entropy gradient starvation framework of Pezeshki et al. [2021].

5.1 Continuous-time limit and timescale separation

We analyze the continuous-time limit of joint SGD, in which the parameters (θ_t, ϕ_t) evolve according to the gradient flow

$$\dot{\theta}(t) = -\nabla_{\theta}\mathcal{L}(\theta, \phi), \quad \dot{\phi}(t) = -\nabla_{\phi}\mathcal{L}(\theta, \phi). \quad (4)$$

This limit is standard under the Robbins-Monro step-size schedule and captures the leading-order dynamics for sufficiently small learning rates. The presence of stochastic gradient noise modifies the constants but not the qualitative behavior Borkar [2008].

Lemma 21 (Effective timescale). *Define the effective timescale ratio*

$$\varepsilon := \frac{\lambda_{\min}(H_{\theta\theta})}{\lambda_{\max}(H_{\phi\phi})}.$$

Under the conditions of Theorem 11, $\varepsilon = \mathcal{O}(C_f/C_g)$ and hence $\varepsilon \rightarrow 0$ in the regime of Assumption 5.

Proof. Immediate from Lemmas 12 and 16. □

After rescaling, the gradient flow (4) takes the form of a singularly perturbed system

$$\dot{\phi}(t) = -\nabla_{\phi}\mathcal{L}(\theta, \phi), \quad \dot{\theta}(t) = -\varepsilon \cdot \nabla_{\theta}^{\text{rescaled}}\mathcal{L}(\theta, \phi), \quad (5)$$

where $\nabla_{\theta}^{\text{rescaled}}$ denotes the gradient in the geometry induced by the spectral Hessian block.

5.2 Fast equilibrium manifold

Under (5), ϕ moves on a fast timescale and θ on a slow one. The classical singular perturbation theorem of Tikhonov (see Borkar [2008], Chapter 6, and Heusel et al. [2017] for the machine-learning adaptation) implies that ϕ converges rapidly to a quasi-equilibrium manifold parametrized by θ .

Lemma 22 (Fast equilibrium). *Under (5) and the regularity Assumption 3, there exists a smooth manifold $\phi^* : \mathbb{R}^{C_f} \rightarrow \mathbb{R}^{C_g}$ such that for any fixed θ , the fast dynamics $\dot{\phi} = -\nabla_{\phi}\mathcal{L}(\theta, \phi)$ converge to $\phi^*(\theta)$ in time $\mathcal{O}(1)$.*

Remark 23. $\phi^*(\theta)$ is the configuration of the spatial model that best classifies the data given the current (and from the spatial model’s view, frozen) spectral features. Importantly, the cross-entropy saturation property described below operates *on this manifold*, not in an asymptotic limit.

5.3 Cross-entropy saturation and residual decay

The crucial dynamical step is that, along the fast manifold $\phi^*(\theta)$, the cross-entropy residual decays exponentially fast. This is the gradient starvation phenomenon of Pezeshki et al. Pezeshki et al. [2021] specialized to our compositional setting.

Lemma 24 (Residual decay along the fast manifold). *Let $r(t) := \text{softmax}(F_{\theta, \phi^*(\theta)}(X)) - y_{\text{onehot}}$ denote the per-pixel cross-entropy residual under the fast equilibrium. Assume the spatial model has sufficient capacity to interpolate the training data using any feature map that contains a sufficient spatial signal—a condition we make precise in the proof. Then*

$$\|r(t)\|^2 \leq C \cdot e^{-\mu t}$$

for constants $C, \mu > 0$ depending on the data and architecture.

Remark 25. The proof of Lemma 24 closely follows the NTK analysis of Pezeshki et al. [2021] for cross-entropy losses. The key difference in our setting is that the dominant features along which the residual decays are spatially-encoded rather than learned-spectrally: the spatial model, having enough capacity to exploit positional and contextual structure in the input, achieves low training loss without requiring the spectral module to specialize. [TODO: Phase 3, W10: write the NTK-based residual decay proof in full, adapting Pezeshki Theorem 3.1 to the compositional setting. Verify the spatial-signal sufficiency condition holds in our synthetic experiments.]

5.4 Main result

Theorem 26 (Two-Timescale Spurious Convergence). *Under Assumptions 1–5 and the spatial-signal sufficiency condition of Lemma 24, the joint gradient flow (4) converges to a point (θ^*, ϕ^*) satisfying*

$$\nabla \mathcal{L}(\theta^*, \phi^*) = 0,$$

and

$$\|\theta^* - \theta_0\| = \mathcal{O}(\varepsilon),$$

where θ_0 is the initialization. In particular, the spectral module is asymptotically pinned to a small neighborhood of its initialization, independently of the labels.

Proof sketch. By Lemma 22, after time T_{fast} of order $\mathcal{O}(1)$, the fast variable ϕ has converged to $\phi^*(\theta_0)$. By Lemma 24, along this manifold the residual r decays exponentially: $\|r(t)\| \leq C e^{-\mu t}$.

The slow-variable update (5) satisfies

$$\|\dot{\theta}(t)\| = \varepsilon \cdot \|J_{\theta}^{\top} r(t)\| \leq \varepsilon \cdot \|J_{\theta}\|_{\text{op}} \cdot C \cdot e^{-\mu t}.$$

Integrating from T_{fast} to ∞ ,

$$\|\theta^* - \theta_0\| \leq \varepsilon \cdot \|J_{\theta}\|_{\text{op}} \cdot C \cdot \frac{e^{-\mu T_{\text{fast}}}}{\mu} = \mathcal{O}(\varepsilon).$$

The fixed-point condition $\nabla \mathcal{L} = 0$ holds because both $r \rightarrow 0$ (so $\nabla_{\theta} \mathcal{L} \rightarrow 0$) and ϕ is at the fast equilibrium ($\nabla_{\phi} \mathcal{L} = 0$ on the manifold). $\square$

5.5 Interpretation

Theorem 26 formalizes the central claim of the paper. The point (θ^*, ϕ^*) is mathematically a critical point of the loss but is, in any practical sense, a failure: the spectral parameters have not departed materially from their initialization, and the spatial module has interpolated the labels using whatever spatial signal exists in the (near-random or pretrained) feature map. Whether the spectral module *could have learned* useful features under freezing-then-decoupling or other interventions is precisely what our prescriptive remedy (Section 9) addresses.

5.6 Comments on the assumptions

The spatial-signal sufficiency condition deserves comment. We require that the spatial model can interpolate the training labels using only the structure of the feature map Z (e.g., positional cues, contextual patterns), without requiring f_θ to specialize. In practice this condition is satisfied whenever the spatial model has sufficient capacity to overfit the training set—which is the norm for modern overparameterized ViTs and CNNs. We verify the condition empirically in Experiments 1.2 and 1.3 by showing that joint training reaches near-zero training loss with a spectral module that has not appreciably changed from initialization.

[TODO: Phase 3, W10: Formalize the spatial-signal sufficiency condition. Likely route: bound on Rademacher complexity of $g_\phi \circ Z$ that guarantees zero-error interpolability without requiring rank growth in Z . Cross-check against Pezeshki et al. [2021] Section 3.]

6 The Effective Gradient Ratio Diagnostic

Theorems 11 and 26 together identify a training pathology that is invisible to standard loss curves: as cross-entropy decays toward zero on both training and validation sets, the spectral module is silently gradient-starved. A practitioner running joint training sees the loss go down and the validation accuracy go up, with no in-training signal that the spectral module has failed to specialize. The implication for held-out test performance is only revealed once test data is examined—often after many epochs of wasted compute, or after deployment.

This section introduces the *Effective Gradient Ratio* (EGR), a real-time diagnostic that detects gradient starvation as it occurs. EGR is a single scalar, computable inside any training loop at negligible cost, that converts the asymptotic prediction of Theorem 26 into a concrete observable.

6.1 Definition and basic properties

Recall from Definition 8 that at training step t ,

$$\text{EGR}(t) = \frac{\|\nabla_{\theta} \mathcal{L}(\theta_t, \phi_t)\|}{\|\nabla_{\phi} \mathcal{L}(\theta_t, \phi_t)\|}.$$

The numerator and denominator are computable directly from gradients accumulated during the standard backward pass. EGR is unitless and invariant to the choice of learning rate.

Proposition 27 (Initial value). *At initialization, under random Gaussian weights with variance scaled in the standard way (e.g. Kaiming initialization),*

$$\mathbb{E}[\text{EGR}(0)] = \Theta\left(\sqrt{C_f/C_g}\right).$$

Remark 28. Proposition 27 captures the intuition that at initialization, the gradient norms reflect the parameter counts of each module: $\|\nabla_{\phi} \mathcal{L}\|^2 \sim C_g$ in expectation while $\|\nabla_{\theta} \mathcal{L}\|^2 \sim C_f$. The square root arises from the norm. For our motivating architecture with $C_g/C_f \approx 325$, this gives $\mathbb{E}[\text{EGR}(0)] \approx 0.055$. [TODO: Phase 4, W11: derive the exact constants under Kaiming initialization. Verify empirically in Experiment 1.1.]

6.2 Monotonicity along the slow-fast manifold

Proposition 29 (EGR decay). *Under the conditions of Theorem 26, along the fast equilibrium manifold $\phi = \phi^*(\theta)$,*

$$\text{EGR}(t) \leq \text{EGR}(t_0) \cdot \exp(-\mu(t - t_0))$$

for $t \geq t_0 \geq T_{\text{fast}}$, where μ is the residual decay rate from Lemma 24.

Proof sketch. By the chain-rule decomposition (2),

$$\|\nabla_{\theta}\mathcal{L}\| \leq \|\partial\hat{y}/\partial Z\|_{\text{op}} \cdot \|\partial Z/\partial\theta\|_{\text{op}} \cdot \|r\|.$$

On the fast manifold, the residual r decays as $\|r\| \leq Ce^{-\mu t}$ by Lemma 24, while $\|\nabla_{\phi}\mathcal{L}\|$ also depends on r but is regularized by the spatial Hessian curvature, which under Theorem 11 scales with C_g . The ratio therefore decays at rate $\Omega(\mu)$. [TODO: Phase 4, W11: tighten the proof to give exact constants on the denominator’s decay rate.] $\square$

6.3 Threshold criterion

Proposition 29 suggests a practical criterion for detecting the onset of gradient starvation.

Definition 30 (Gradient starvation alert). We say the model is in a *gradient starvation regime* at step t if

$$\text{EGR}(t) < \tau \cdot \text{EGR}(0),$$

where $\tau \in (0, 1)$ is a user-supplied tolerance. Empirically, we recommend $\tau \in [0.05, 0.1]$, corresponding to a 10–20 $\times$ drop in EGR relative to initialization.

Remark 31. The threshold τ is the only hyperparameter of the diagnostic. The default $\tau = 0.1$ corresponds to a one-decade decay and is the value we use throughout Section 7. We provide a sensitivity analysis to τ in the supplement.

Practical use. A practitioner logs $\text{EGR}(t)$ at each step. When the alert fires, two options are available:

1. **Freeze the spectral module.** This is the prescription most directly motivated by Theorem 26: removing θ from the optimization eliminates the timescale gap. We discuss this in Section 9.
2. **Inject a fixed residual baseline.** A softer intervention: replace $Z = f_{\theta}(X)$ with $Z = f_{\theta}(X) + h(X)$ for a fixed feature extractor h (e.g. PCA). The fixed baseline ensures the spatial model receives stable input even if f_{θ} continues to drift, breaking the shortcut without freezing.

We evaluate both interventions in Experiment 1.3.

6.4 Relationship to existing diagnostics

EGR is conceptually related to several existing notions but is distinct in its scope and intent.

GradNorm Chen et al. [2018]. GradNorm dynamically reweights losses in multi-task learning to equalize gradient magnitudes across tasks. EGR measures rather than intervenes, and applies to single-task compositional architectures rather than multi-task losses.

Modality-specific learning rates (MSLR). In multimodal learning, MSLR adjusts learning rates per modality. EGR diagnoses the underlying pathology that MSLR addresses but in a unimodal, two-stage architecture where the “modalities” are the spectral and spatial subspaces of a single input.

Loss curve monitoring. The conventional practice of monitoring training and validation loss fails to detect gradient starvation because both can decay even as θ stagnates. EGR is sensitive to the structural failure even when no global loss curve abnormality is present.

6.5 Empirical validation: EGR as a diagnostic

We validate EGR as a diagnostic of training pathology in two stages. First, we establish that EGR statistics correlate with generalisation across our full sweep. Second, we test whether the signal is genuine or merely a re-encoding of model capacity.

Pooled correlation (capacity confounded). Across $N = 24$ runs spanning four widths $\{16, 64, 256, 1024\}$ with six seeds each, EGR depth correlates strongly with final test accuracy: Pearson $r = -0.723$ ($p = 6.6 \times 10^{-5}$), Spearman $\rho = -0.808$ ($p = 1.8 \times 10^{-6}$), and with the train-test gap at $r = +0.741$ ($p = 3.5 \times 10^{-5}$). However, EGR depth is itself strongly correlated with log width ($r = +0.903$): roughly 77% of the pooled correlation with test accuracy is attributable to capacity rather than to EGR depth per se (partial r controlling for log width collapses to -0.169 , $p = 0.44$). The pooled correlation alone cannot establish a capacity-independent diagnostic.

Fixed-capacity test. To isolate the capacity-independent signal, we ran a second experiment holding CNN width fixed at $D = 256$ and varying data noise $\{0.05, 0.10, 0.15, 0.20\}$, learning rate $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, weight decay $\{0, 10^{-3}\}$, and six seeds, for a total of $N = 144$ runs. At fixed capacity, $\text{EGR}_{\min}$ is the relevant statistic (with width fixed, `egr_depth` reduces to a constant offset of `egr_min` in expectation). It exhibits a within-capacity correlation with generalisation that is statistically robust but practically moderate: Pearson $r(\text{EGR}_{\min}, \text{acc}) = +0.344$ ($p = 2.5 \times 10^{-5}$).

The relationship survives every adversarial control we applied. $\text{EGR}_{\min}$ is essentially orthogonal to the swept hyperparameters — marginal correlations with noise, learning rate, and weight decay are $+0.029$, -0.073 , and $+0.007$ respectively, with the full hyperparameter set explaining only $R^2 = 0.015$ of $\text{EGR}_{\min}$ variance. The partial correlation $r(\text{EGR}_{\min}, \text{acc} \mid \text{noise}, \text{lr}, \text{wd}) = +0.395$ ($p = 9.5 \times 10^{-7}$) is *larger* than the pooled $r = +0.334$, which is the opposite signature of a hidden hyperparameter confound. The within-bin analysis is the strongest evidence: across the 24 (noise, lr, wd) cells in which only the seed varies, 23 of 24 yield strongly positive $r(\text{EGR}_{\min}, \text{acc})$ (mean $+0.584$, Fisher- z aggregated $t = 14.65$, $p < 10^{-13}$).

Controlling for final training loss — itself a non-trivial predictor with $r = +0.47$ for accuracy and $r = -0.55$ for the overfit gap — leaves the $\text{EGR}_{\min}$ partial correlation essentially unchanged at $+0.335$ ($p = 4.1 \times 10^{-5}$). EGR therefore carries information that train-loss monitoring does not.

Practical interpretation. Honesty about effect size is required. Treated as a binary alarm, $\text{EGR}_{\min}$ achieves ROC-AUC = 0.612 for flagging below-median test accuracy, and $\text{EGR}_{\text{depth}}$ achieves only 0.538 for high overfit gap. The best operating point ($\text{EGR}_{\min} < 0.10$) catches 90% of bad runs but also flags 57% of healthy runs — no threshold delivers a usable trade-off as a standalone indicator.

Verdict. At fixed capacity, EGR is a real, capacity-independent signal correlated with generalisation. The within-capacity effect size is small enough that EGR is best deployed alongside train-loss monitoring as a complementary diagnostic, not as a standalone run-quality indicator. For research and ablation work, EGR provides genuine insight into the gradient starvation mechanism beyond what loss monitoring reveals; for production monitoring of single training runs, plain loss curves remain the primary signal and EGR a useful adjunct.

6.6 Implementation

The EGR diagnostic is implemented as a PyTorch training callback. It hooks into the standard training loop between the backward pass and the optimizer step:

```

loss.backward()
egr_logger.log_step(global_step)    # <-- the only added line
optimizer.step()
optimizer.zero_grad()

```

The callback identifies the spectral and spatial parameter groups via `model.spectral.parameters()` and `model.spatial.parameters()` (or any user-supplied naming convention) and records the gradient norms and EGR at each step. We release the implementation as the `egr` package, available at the project repository. **[TODO: Phase 4, W12: prepare PyPI package skeleton. Add unit tests for gradient norm computation and threshold detection.]**

7 Synthetic Experiments

We validate Theorems 11 and 26 and the EGR diagnostic on a controlled synthetic spectral-spatial classification problem. Three experiments isolate distinct theoretical predictions: (1.1) the Hessian condition number bound; (1.2) the two-timescale dynamics under joint training; (1.3) the equal-information killer test demonstrating that joint training prefers spatial features even when spectral and spatial signals carry equal information.

7.1 Synthetic data and architectures

We use a calibrated generator of synthetic hyperspectral images $X \in \mathbb{R}^{H \times W \times S}$ with $H = W = 16$ and $S = 64$, supporting two-class per-pixel classification with controlled spectral and spatial signal strengths through scalar parameters α and β . The data generator places a known spectral direction $u \in \mathbb{R}^S$ in each pixel scaled by a per-pixel latent $z \sim \mathcal{N}(0, 1)$ and modulated by α , with smooth per-class spatial templates modulated by β entering the label-generating logits. Bayes-optimal accuracy is ≈ 0.85 – 0.95 at $\alpha = \beta = 1$ and noise level 0.1.

For the spatial module g_ϕ we use three architectures: a two-layer convolutional network (SpatialCNN, 3×3 kernels), a two-layer pre-LN Vision Transformer (SpatialViT, pixel-token), and a per-pixel MLP baseline (SpatialMLP, no spatial receptive field). The MLP is included for ablation; it cannot exhibit a spatial shortcut by construction.

7.2 Experiment 1.1: Hessian capacity ablation

We vary the spatial width D across two orders of magnitude ($D \in \{16, 32, 64, \dots, 8192\}$ for SpatialMLP, equivalent parameter scales for other architectures) with $C_f = 1024$ held fixed. For each (D, seed) we compute the top eigenvalue of the spectral and spatial block Gauss-Newton matrices via power iteration (5 seeds per width). The predictions to be tested are $\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(C_g)$ (Lemma 12) and $\lambda_{\max}(J_\theta^\top J_\theta)$ approximately constant (Lemma 16).

Findings. Across 50 measurements:

- $\lambda_{\max}(J_\theta^\top J_\theta)$ is flat at ≈ 0.02 over the full range of D – the spectral block does not grow with the spatial capacity, consistent with Lemma 16.
- $\lambda_{\max}(J_\phi^\top J_\phi)$ grows by $74\times$ (from 0.62 to 46) as C_g grows by $510\times$.
- The empirical log-log slope of $\lambda_{\max}(J_\phi^\top J_\phi)$ vs. C_g is 0.70 ± 0.05 across the full range. The Karakida prediction is slope 1; we attribute the gap to finite-width effects with the asymptote approached only at much larger widths.
- The condition number $\kappa = \lambda_\phi / \lambda_\theta$ grows from 40 at $C_g/C_f = 0.3$ to 2,407 at $C_g/C_f = 152$. At the capacity ratio of our motivating architecture ($C_g/C_f \approx 325$), this extrapolates to κ in the thousands.

The qualitative geometric prediction is verified: the spectral and spatial block eigenvalues diverge with capacity, producing a severely ill-conditioned joint loss landscape. The finite-size scaling exponent is somewhat smaller than the asymptotic linear prediction, which we discuss in the limitations.

Headline figure. Figure ?? (paired view: λ_ϕ and λ_θ on the same axes across the D sweep) shows the two eigenvalues diverging as D grows. The constancy of λ_θ is itself a strong qualitative finding: the spectral block carries no growing curvature with D , isolating the asymmetry to the spatial side.

7.3 Experiment 1.2: Two-timescale training dynamics

We train the compositional model under joint and frozen-spectral conditions across multiple capacities and architectures, logging the EGR every step. The predictions to be tested are Proposition 29 (EGR decays along the slow-fast manifold) and the consequence that the EGR collapse depth should scale monotonically with capacity.

Findings — CNN. On SpatialCNN at widths $D \in \{16, 64, 256, 1024\}$ with 3 seeds each:

- EGR collapse depth is monotonic in D :

$$\text{EGR}_{D=16}^{\text{final}} \approx 0.40, \quad \text{EGR}_{D=64}^{\text{final}} \approx 0.25, \quad \text{EGR}_{D=256}^{\text{final}} \approx 0.10, \quad \text{EGR}_{D=1024}^{\text{final}} \approx 0.08.$$

- Joint training final test loss diverges at high capacity. At $D = 1024$, joint test loss = 2.27 while frozen test loss = 1.15.
- Joint test accuracy peaks early (≈ 0.70) then decays as the model overfits to shortcuts. Frozen test accuracy is more stable.

This is the cleanest empirical validation of Theorem 26: on the canonical CNN architecture, EGR exhibits exactly the capacity-monotonic exponential decay predicted by Proposition 29.

Findings – ViT. On SpatialViT, training enters a full memorization regime: train loss reaches 0.01–0.03 at all widths, while test loss is catastrophic (2.5–5.0). In this regime the EGR observable loses interpretive value: both gradient norms approach the noise floor and their ratio is dominated by stochasticity. The underlying prescription, however, still holds: frozen variants beat joint variants at every width tested.

Conclusion. Theorem 26’s qualitative prediction (joint training fails by undertraining the spectral module) is universal: it manifests on both CNN and ViT spatial models. The specific EGR observable cleanly tracks the predicted decay in non-memorization regimes (CNN at our capacities) and becomes uninformative in the memorization regime (ViT). We discuss this limitation in Section 9.

7.4 Experiment 1.3: The equal-information killer test

The previous experiments validate the geometric (Theorem 11) and dynamical (Theorem 26) predictions on capacity-asymmetric compositional models. Experiment 1.3 tests a stronger qualitative claim suggested by the theory: *when spectral and spatial signals carry equal information about the label, end-to-end joint training will underperform freezing because of the shortcut dynamics.*

Calibration. We require an operational definition of “equal information” between the spectral pathway (per-pixel content of X along a direction $u \in \mathbb{R}^S$) and the spatial pathway (per-pixel content of X along an orthogonal direction $v \perp u$, with strength modulated by a smooth per-position function of the class). We compare three modes:

- *bayes*: scalar strengths α, β chosen such that a single-modality logistic regression on α -only data and a position-majority classifier on β -only data both achieve a target accuracy (0.75 in our experiments);
- *ntk*: α, β chosen such that the leading Gram-matrix eigenvalues of the per-pixel features from the two modalities are equal;
- *margin*: α, β chosen such that the L2 margins of single-modality logistic regressions are equal.

Setup. For each calibration mode we train the canonical CNN composition at width $D = 256$ for 150 epochs on $N_{\text{train}} = 512$ samples, repeated over 3 seeds. The four training conditions per mode are:

- *joint*: end-to-end training with both modules trainable;
- *frozen*: spectral module frozen at random initialization;
- *spectral-only*: $\beta = 0$, only spectral signal in X ;
- *spatial-only*: $\alpha = 0$, only spatial signal in X .

Results. Under the **bayes** calibration we find:

- The single-modality baselines reach equal accuracy by construction: $\alpha = 0.82$, $\beta = 18.75$ produces spectral-only = 0.746, spatial-only = 0.754 (logistic regression and position-majority bounds respectively).
- Empirically on the full CNN model: spectral-only training (data with $\beta = 0$) achieves test accuracy 0.53, spatial-only training (data with $\alpha = 0$) achieves test accuracy 0.53. The CNN-achievable accuracies are equal, confirming the calibration is meaningful in architecture.
- Frozen-spectral training on the full data achieves 0.61, exploiting both modalities.
- **Joint training catastrophically collapses to 0.48** – below chance, below either single-modality baseline, and 0.13 points below the frozen variant.

The collapse below the spectral-only baseline is the strongest result. End-to-end training does not merely fail to combine the two modalities – it actively destroys the spectral information that the model demonstrably can use when trained on spectral-only data. This is the spectral-shortcut prediction made empirically falsifiable.

Comparison of calibration modes. Table 1 compares the three calibration modes on two criteria: (i) calibration quality, defined as $1 - |a_{\text{spec}} - a_{\text{spat}}|$ where $a_{\text{spec}}, a_{\text{spat}}$ are the achieved single-modality accuracies; and (ii) shortcut strength, defined as $a_{\text{frozen}} - a_{\text{joint}}$.

Conclusion. The **bayes** calibration delivers the qualitative claim: when spectral and spatial signals carry equal information about the label (measured by what each modality alone can provide), end-to-end joint training collapses below the better single-modality baseline. Freezing the spectral module recovers the information that joint training squanders. This is the direct empirical demonstration of the spectral-shortcut conjecture suggested by Theorem 26.

mode	α, β	spec / spat	joint	frozen	shortcut
bayes	0.82, 18.75	0.75/0.75	0.48	0.61	+0.13
ntk	0.99, 4.91	0.78/0.58	0.53	0.57	+0.04
margin	5.00, 5.00	0.95/0.59	0.60	0.69	+0.09

Table 1: Calibration mode comparison for the equal-information killer test (CNN $D = 256$, 3 seeds). *spec / spat*: achievable single-modality accuracies. *shortcut* $:= a_{\text{frozen}} - a_{\text{joint}}$. The **bayes** mode produces both the cleanest modality equalization and the largest shortcut effect.

7.5 Robustness

In supplementary experiments we vary the optimizer (Adam vs. SGD with momentum), the noise level (0.05 vs. 0.10), and the spectral input dimension S . The capacity-driven joint-vs-frozen gap and the EGR collapse pattern hold across all variations, with two notable modulations: SGD makes the EGR signal cleaner (Adam normalizes per-parameter and partially masks the natural gradient ratio), and lower noise allows cross-entropy to fully saturate, accelerating the emergence of the shortcut.

[TODO: Phase 5, W15: write the robustness analyses into the supplement once Exp 1.3 is finalized.]

8 Real-Data Validation

The synthetic experiments in Section 7 verify the theorems and the EGR diagnostic in a controlled setting. We here briefly confirm that the predicted phenomenology manifests in a production-scale spectral-spatial classification task: per-pixel 4-class tissue classification on FTIR/QCL infrared hyperspectral images of breast cancer tissue microarrays. A companion empirical paper Dziuba et al. [2026] provides comprehensive ablations on this dataset and on a parallel prostate cancer dataset.

8.1 Setup

The data are 336×336 pixel cores collected by FTIR/QCL spectroscopy over the $1000\text{--}1800\text{ cm}^{-1}$ fingerprint region. Each pixel carries an $S = 942$ dimensional spectrum (after stacking raw absorbance and its first two derivatives). The classification target is per-pixel tissue type. The compositional model uses a linear $f_{\theta} : \mathbb{R}^{942} \rightarrow \mathbb{R}^{128}$ followed by a 12-layer Vision Transformer plus ConvTranspose decoder. Parameter counts: $C_f = 120,576$, $C_g \approx 13 \times 10^6$; capacity ratio $C_g/C_f \approx 325$. Per Theorem 11, this predicts κ of order $325 \cdot c_1$ where c_1 is bounded below by the constants of Lemmas 12–16.

We evaluate five model variants under identical 5-fold cross-validation splits: end-to-end learned linear (joint), frozen random spectral, frozen PCA-128 spectral, frozen pretrained spectral encoder (pretrained on the pixel-level classification task before downstream training), and fine-tuned pretrained spectral (initialize from the pretrained encoder, then continue with end-to-end training).

8.2 Results

The mean test F1 scores across the 5 folds are summarized in Table 2.

Three observations align directly with the theory:

1. **Joint training loses to frozen-random.** End-to-end learned linear achieves 0.675, while frozen random spectral features achieve 0.70. This is the strongest single datapoint for

Variant	Test F1
End-to-end learned linear (joint)	0.675
Frozen random spectral	0.70
Frozen PCA-128 spectral	0.78
Frozen pretrained spectral (Peak Windows)	0.84
Frozen pretrained spectral (MLP)	0.90
Frozen pretrained spectral (Sliding Win)	0.95
Fine-tuned pretrained spectral	0.79

Table 2: Test F1 on held-out FTIR/QCL breast cancer tissue cores (5-fold mean). End-to-end learned linear collapses to 0.675, while frozen random features outperform it ($0.70 > 0.675$). Frozen pretrained variants achieve 0.84–0.95. Fine-tuning the pretrained encoder degrades performance from 0.90 to 0.79.

Theorem 26: even uninformative features, when frozen, outperform supervised learning of the same spectral module.

2. **Frozen PCA improves over frozen random.** Adding informativeness (PCA) on top of stability (frozen) improves test F1 by 0.08. The benefit is monotone in feature informativeness, and the gap is preserved because the spatial model receives a non-moving target.
3. **Fine-tuning the pretrained encoder degrades it.** Starting from a strong frozen pretrained (0.90) and allowing the gradient to flow through it for fine-tuning drops performance to 0.79. This is the predicted dynamical consequence of Theorem 26: the same shortcut dynamics that prevent end-to-end training from succeeding also actively degrade pretrained features when they are allowed to update.

The ordering of variants is, top to bottom in Table 2, exactly what the theory predicts: any frozen variant beats any joint or fine-tuned variant; informativeness of the frozen features sets the final achievable level; joint and fine-tuned training is uniformly worse.

8.3 Diagnostic on real data

[TODO: Phase 6, W16-17. Compute EGR trajectories for each of the 5 variants during the original 5-fold training runs. Predict: learned linear shows early EGR collapse, frozen variants have undefined EGR (theta has no gradient), fine-tuned shows mid-training EGR collapse as the pretrained features are dragged off the manifold. Plot in a single figure with all variants overlaid.]

The companion empirical paper Dziuba et al. [2026] provides additional ablations: capacity ratio sweeps, prostate cancer cohort replication, train/val/test split sensitivity analysis. The numerical results across these ablations all conform to the predicted theory.

9 Discussion

9.1 Implications for practice

The theorems and experiments establish a concrete prescription. In any compositional spectral-spatial pipeline with substantial capacity asymmetry $C_g \gg C_f$, end-to-end joint training is expected to fail in a specific and predictable way: the spectral module stays near initialization while the spatial module fits the data using shortcuts in the unstructured feature map. Three operational responses follow.

Freeze the spectral module. The most direct response to Theorem 26 is to remove θ from the optimization. Without a spectral update, the timescale separation is eliminated by construction, and g_ϕ optimizes a standard supervised learning problem on fixed input features. If a pretrained spectral encoder is available (from unsupervised pretraining, self-supervised pretraining, or pre-task pretraining on the same data), it should be used frozen.

Use a fixed baseline. When pretraining is unavailable, a fixed but informative baseline $h(X)$ (e.g. PCA, fixed wavelet basis, random projection) can be injected as an additive component $Z = f_\theta(X) + h(X)$. The fixed component ensures the spatial model receives a stable input even if f_θ drifts; the additive structure allows fine-grained refinement of f_θ without exposing the spatial model to the moving-target dynamics that produce the shortcut. We have not empirically tested this prescription in this paper; it follows directly from the theory and is a natural focus for follow-up work.

Track EGR. The EGR diagnostic detects the onset of starvation in real time. Practitioners should log it alongside training and validation loss. When $\text{EGR}(t)$ drops below a chosen threshold (we recommend $0.1 \cdot \text{EGR}(0)$), the theoretical analysis predicts that subsequent θ updates are essentially noise: the spectral module has been absorbed into a spurious stationary point. At this point, freezing θ and continuing to train ϕ retains the run’s investment while avoiding the negative consequences of further joint updates.

9.2 Limitations

We make four limitations explicit.

Linear spectral reduction. Theorems 11 and 26 are proven under the assumption that f_θ is linear in θ . In practice some pipelines use shallow nonlinear spectral modules (1–2 layer MLPs). The intuition behind both theorems extends: nonlinear shallow f_θ retains the property that the spectral block carries substantially less capacity than the spatial block, so the qualitative timescale separation persists. We verify empirically in Section 7 that the joint-vs-frozen gap remains under nonlinear f_θ . A formal proof for nonlinear f_θ is left to future work.

Finite-width scaling exponent. The empirical scaling exponent of λ_ϕ vs. C_g is ~ 0.7 at the widths we tested, while the asymptotic Karakida prediction is 1.0. We expect the asymptote to be approached at larger widths. Two consequences for the paper: the Theorem 11 bound should be read as a lower bound that may be loose at finite scale, and the empirical predictions should be interpreted as qualitative (monotonic growth) rather than tight (constant proportionality).

Memorization regime. For sufficiently overparametrized spatial models (ViT in our experiments), joint training enters a memorization regime in which train loss falls to near zero and both gradient norms approach the noise floor. In this regime the EGR observable becomes uninformative, even though the underlying prescription (freeze) continues to apply. Practitioners running highly overparametrized models should observe the symptoms (test loss divergence, train loss saturation) rather than relying on EGR alone.

Single domain validation. Our real-data validation is restricted to FTIR/QCL tissue classification in breast cancer. We expect the same mechanism to operate in hyperspectral remote sensing, Raman tissue classification, and mass spectrometry imaging, but those validations are future work.

9.3 Defense against the "linear triviality" critique

A natural reviewer objection is that, because f_{θ} is linear, the analysis must reduce to principal component analysis. We address this in Remark 20 and emphasize again: PCA finds directions of maximum input variance independently of labels. Theorem 11 concerns curvature of the supervised cross-entropy loss, which depends on the current state of g_{ϕ} through the Jacobian $\partial \hat{y} / \partial Z$. The pathology we identify is that this supervised signal fails to reach θ . A PCA-decomposed f_{θ} would not exhibit the joint-vs-frozen gap we observe: indeed, our experiments include a frozen-PCA baseline that performs well, exactly because PCA decomposition does not require the supervised signal to flow through it.

9.4 The Spatial Dominance Conjecture: an open problem

A stronger statement than Theorem 26 is suggested by our experiments but not proven: *when spectral and spatial features carry equal information about the label, joint training exhibits an implicit bias toward the spatial pathway*. Pezeshki et al. [2021] prove a related but distinct result in the NTK regime; Huang et al. [2022] prove a related result for explicit dual-modality networks but not for compositional single-modality pipelines. A proof of the spatial-dominance conjecture in our setting would require formalizing “equal information” in an operationally meaningful way (Shannon mutual information is bijection-invariant and therefore not the right primitive for gradient dynamics) and combining the NTK eigenvalue gap with our compositional Hessian analysis. We leave this as an open problem.

9.5 Beyond spectroscopy: where else this matters

Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We highlight three areas where the same pathology likely operates.

Hyperspectral remote sensing. The dominant paradigm for hyperspectral image classification is a shallow spectral reduction (PCA, autoencoder, or 1D conv) feeding into a deep spatial classifier (3D CNN, ViT). The capacity asymmetry is typically large. Joint training is the de facto standard. Our analysis predicts the same shortcut emergence; whether the field has noticed it under the guise of “classification works without much spectral processing” is a question worth examining.

Multimodal foundation models. Vision-language and audio-language models compose a sequence of modality-specific encoders into a shared representation space. The modality with the larger encoder typically dominates the shared representation. While this has been documented empirically (Wang et al. [2020], Peng et al. [2022]), the mechanism is the same as ours: capacity-induced timescale separation combined with cross-entropy saturation. Our framework extends their empirical observations into the single-input compositional setting.

Video models with frozen image encoders. A standard practice in video understanding is to freeze a pretrained image encoder and train a temporal model on top. This is widely treated as “transfer learning,” but the same mechanism we describe predicts that joint training would fail in a specific way – the image encoder would drift, the temporal model would absorb the shortcut, generalization would suffer. The empirical literature strongly supports the freezing practice; our analysis provides a mechanistic reason.

9.6 Concluding remarks

We have provided the first rigorous explanation for the failure of end-to-end joint training in compositional spectral-spatial deep learning. The phenomenon is geometric (Theorem 11) and dynamical (Theorem 26), is detectable in real time via the EGR diagnostic, and has a direct prescriptive remedy (freezing or fixed-baseline injection). We expect the framework to extend beyond spectroscopy to any compositional architecture with capacity asymmetry, and to provide a principled foundation for the empirical practice of frozen feature extractors that the field has converged on by necessity rather than design.

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A Supplementary Material

This supplement provides the full proofs of Theorems 11 and 26, additional experiments referenced in the main text, and robustness checks.

A.1 Full proof of Theorem 11

We restate Theorem 11 for reference:

Under Assumptions 1-6, the effective condition number of the joint Gauss-Newton matrix satisfies $\kappa_{\text{eff}}(G) \geq c_1 \cdot \sqrt{C_g/C_f} - c_2$ for some $c_1, c_2 > 0$ depending on the data covariance, the bottleneck dimension K , and the model class of g_ϕ .

The proof is structured as three lemmas: (i) the spatial block’s top eigenvalue scales with $\sqrt{C_g}$ (Lemma 12, Karakida-style mean-field FIM analysis); (ii) Cauchy interlacing transfers this scaling to the full Hessian (Lemma 14); (iii) a residual-dimension rank bound caps the smallest positive eigenvalue by a C_g -independent constant (Lemma 16). We prove each in turn, then combine.

A.1.1 Proof of Lemma 12

We show $\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(C_g)$ in expectation over random Gaussian initialization, under Assumptions 3 (twice-differentiable, L -Lipschitz gradients) and 4 (sub-Gaussian inputs with bounded covariance condition number).

Step 1: Bounded per-pixel residual at initialization. Under Kaiming initialization, each pre-activation in the spatial module g_ϕ has $\mathcal{O}(1)$ variance. The output logits $\hat{y} = g_\phi(f_\theta(X))$ are therefore distributed as $\hat{y} \sim \mathcal{N}(0, \sigma_y^2 I_{N_{cls}})$ at initialization in the mean-field limit, with $\sigma_y^2 = \Theta(1)$ independent of C_g . The softmax probabilities $p = \text{softmax}(\hat{y})$ are concentrated near uniform $1/N_{cls}$, and the per-pixel residual $r = p - y_{\text{onehot}}$ satisfies

$$\mathbb{E}[\|r\|^2] = \frac{N_{cls} - 1}{N_{cls}} + \mathcal{O}(\sigma_y^2) = \Theta(1).$$

This bound is independent of C_g .

Step 2: Trace of the Fisher Information block. Karakida et al. [2019], Theorem 3.1, give that for a deep network at random Kaiming initialization, the expected trace of the per-layer FIM block scales linearly with the layer’s parameter count:

$$\mathbb{E}[\text{tr}(F_\phi)] = \sum_{\ell=1}^L q_\ell \cdot \nu_\ell \cdot M_\ell,$$

where M_ℓ is the parameter count of layer ℓ , q_ℓ is the squared activation magnitude at layer ℓ , and ν_ℓ is the squared backpropagated signal at layer ℓ . Under Kaiming init, $q_\ell, \nu_\ell = \Theta(1)$ uniformly in ℓ , so

$$\mathbb{E}[\text{tr}(F_\phi)] = \Theta\left(\sum_{\ell} M_\ell\right) = \Theta(C_g).$$

Step 3: From Fisher to Gauss-Newton. For softmax cross-entropy, the FIM and Gauss-Newton matrix differ by a factor depending on the prediction distribution:

$$F_\phi = \mathbb{E}_X \left[J_\phi^\top \cdot \text{diag}(p)(I - 1p^\top) \cdot J_\phi \right].$$

At random initialization $p \rightarrow (1/N_{cls})\mathbf{1}$, so the inner matrix has eigenvalues bounded by $1/N_{cls}$. Therefore

$$\text{tr}(F_\phi) \geq \frac{1}{N_{cls}^2} \cdot \text{tr}(J_\phi^\top J_\phi)$$

and combining with Step 2:

$$\mathbb{E} \left[\text{tr}(J_\phi^\top J_\phi) \right] = \Omega(C_g).$$

Step 4: Top eigenvalue from trace via participation ratio. The top eigenvalue is bounded from below by the trace divided by the effective rank:

$$\lambda_{\max}(J_\phi^\top J_\phi) \geq \frac{\text{tr}(J_\phi^\top J_\phi)}{\rho(J_\phi^\top J_\phi)},$$

where ρ is the participation ratio (effective rank under the L^2 measure). Karakida et al. [2019], Theorem 4.2, show that under mean-field assumptions, the participation ratio of the FIM is $\Theta(1)$ uniformly in network width. Combining with Step 3:

$$\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(C_g).$$

Step 5: Concentration in the wide-network limit. Karakida et al. [2019], Theorem 5.1, establish that in the mean-field limit, the top eigenvalue of the FIM concentrates around its expectation with high probability. The deterministic bound therefore holds with probability $1 - o(1)$ as $C_g \rightarrow \infty$.

This completes the proof of Lemma 12. $\square$

Remark 32 (Empirical validation). Karakida’s asymptotic law $\lambda_{\max} = \Theta(M)$ corresponds to a log-log scaling exponent of 0.5 against C_g under $C_g = \Theta(M^2)$. Experiment 1.1 measures an empirical exponent of ≈ 0.70 over C_g spanning three decades ($D \in \{16, \dots, 8192\}$), consistent with the asymptotic $\sqrt{C_g}$ prediction plus a mild finite-width upward correction of the same character observed in Karakida’s own numerical validation (their Figures 2–3, exponent $\in [0.5, 0.8]$ across finite widths). The qualitative claim of Lemma 12— λ_ϕ grows without bound as the spatial module widens—is preserved at all widths we test, and the quantitative slope matches the corrected asymptotic prediction.

A.1.2 Proof of Lemma 14

The claim that $\lambda_{\max}(M) \geq \max\{\lambda_{\max}(A), \lambda_{\max}(D)\}$ for a symmetric block matrix $M = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$ is a direct application of Cauchy’s interlacing theorem to principal submatrices. See Bhatia [1997], Corollary III.1.5.

For the eigenvector v achieving $\lambda_{\max}(A) = v^\top A v$, the extension $\tilde{v} = \begin{pmatrix} v \\ 0 \end{pmatrix}$ has $\tilde{v}^\top M \tilde{v} = v^\top A v$, so $\lambda_{\max}(M) \geq \tilde{v}^\top M \tilde{v} / \|\tilde{v}\|^2 = \lambda_{\max}(A)$. The same argument for D completes the proof. $\square$

A.1.3 Proof of Lemma 16

We establish two claims. Part (i) is a rank bound derived from the softmax simplex constraint, aggregated over the pixel grid and dataset. Part (ii) is an upper bound on the smallest nonzero eigenvalue via a trace-over-rank argument, using Assumption 6 to bound the trace independently of C_g .

Convention on $\lambda_{\min}$. The spectral block $J_{\theta}^{\top} J_{\theta}$ is rank-deficient whenever the residual dimension $N_{\text{data}}HW(N_{\text{cls}} - 1) < C_f$ (which is the over-parameterised regime relevant for large spatial classifiers). We interpret the bound on $\lambda_{\min}(J_{\theta}^{\top} J_{\theta})$ throughout this proof as a bound on the smallest *nonzero* eigenvalue $\lambda_{\min}^+$; the strict smallest eigenvalue is zero and any positive upper bound holds trivially. The condition number κ appearing in Theorem 11 is understood as the effective condition number $\kappa_{\text{eff}}(G) = \lambda_{\max}(G)/\lambda_{\min}^+(G)$ per Remark 10 of Section 3.

Step 1: Per-pixel softmax null-space constraint. For each pixel $i \in \{1, \dots, HW\}$ and each training example $n \in \{1, \dots, N_{\text{data}}\}$, the softmax output $p_{n,i}(\theta, \phi) \in \Delta^{N_{\text{cls}}-1}$ satisfies $\mathbf{1}^{\top} p_{n,i} \equiv 1$ identically in θ . Differentiating this identity in θ ,

$$\mathbf{1}^{\top} \frac{\partial p_{n,i}}{\partial \theta} = \mathbf{0}.$$

The one-hot target $y_{n,i}$ also satisfies $\mathbf{1}^{\top} y_{n,i} = 1$, so the residual $r_{n,i} := p_{n,i} - y_{n,i}$ inherits the same null-space constraint: $\mathbf{1}^{\top} r_{n,i} \equiv 0$ and $\mathbf{1}^{\top} \partial r_{n,i} / \partial \theta = \mathbf{0}$. Consequently the per-pixel residual Jacobian block $\partial r_{n,i} / \partial \theta \in \mathbb{R}^{N_{\text{cls}} \times C_f}$ has row-rank at most $N_{\text{cls}} - 1$: one linear dependency between its N_{cls} rows is guaranteed.

Step 2: Aggregation over pixels and dataset. Stack the $N_{\text{data}}HW$ per-pixel Jacobian blocks into

$$J_{\theta} = \frac{\partial r}{\partial \theta} \in \mathbb{R}^{N_{\text{data}}HW N_{\text{cls}} \times C_f}.$$

Each of the $N_{\text{data}}HW$ blocks contributes at most $N_{\text{cls}} - 1$ independent rows (Step 1), and no cross-block dependencies are imposed by the softmax simplex constraint. Hence

$$\text{rank}(J_{\theta}) \leq N_{\text{data}} \cdot HW \cdot (N_{\text{cls}} - 1).$$

Combining with the trivial column-rank bound $\text{rank}(J_{\theta}) \leq C_f$ and the identity $\text{rank}(J_{\theta}^{\top} J_{\theta}) = \text{rank}(J_{\theta})$, we obtain claim (i) of the lemma:

$$\text{rank}(J_{\theta}^{\top} J_{\theta}) \leq \min(C_f, N_{\text{data}}HW(N_{\text{cls}} - 1)). \quad (6)$$

Denote the effective rank ceiling by $R := N_{\text{data}}HW(N_{\text{cls}} - 1)$, which is C_g -independent.

Step 3: Trace bound via encoder-bottleneck chain rule. Under Assumption 1, $\partial Z_i / \partial \theta = I_K \otimes x_i^{\top}$ (Magnus and Neudecker [1988], Ch. 9). Under Assumption 6, $\|\partial \hat{y}_i / \partial Z_i\|_{\text{op}} \leq \tilde{L}$ uniformly. For the softmax residual Jacobian at pixel (n, i) ,

$$\frac{\partial r_{n,i}}{\partial \theta} = [\text{diag}(p_{n,i}) - p_{n,i} p_{n,i}^{\top}] \cdot \frac{\partial \hat{y}_{n,i}}{\partial Z_{n,i}} \cdot \frac{\partial Z_{n,i}}{\partial \theta},$$

with $\|\text{diag}(p) - pp^{\top}\|_{\text{op}} \leq 1$ (the softmax Hessian has operator norm at most 1; see Botev, Ritter, and Barber [2017]). Taking the Frobenius norm,

$$\left\| \frac{\partial r_{n,i}}{\partial \theta} \right\|_F^2 \leq \tilde{L}^2 \cdot \|I_K \otimes x_{n,i}^{\top}\|_F^2 = \tilde{L}^2 \cdot K \cdot \|x_{n,i}\|^2.$$

Aggregating,

$$\text{tr}(J_{\theta}^{\top} J_{\theta}) = \sum_{n,i} \left\| \frac{\partial r_{n,i}}{\partial \theta} \right\|_F^2 \leq \tilde{L}^2 K \cdot \sum_{n,i} \|x_{n,i}\|^2 \leq \tilde{L}^2 K \cdot N_{\text{data}}HW \cdot \text{tr}(\Sigma_X), \quad (7)$$

where Σ_X is bounded by Assumption 4. All factors on the right-hand side are C_g -independent.

Step 4: Trace/rank upper bound on $\lambda_{\min}^+$. For any positive semidefinite matrix A of rank r with eigenvalues $0 = \lambda_1 = \dots = \lambda_{n-r} < \lambda_{n-r+1} \leq \dots \leq \lambda_n$, the arithmetic-mean inequality on the nonzero eigenvalues gives $\lambda_{n-r+1} \leq \text{tr}(A)/r$. Applied to $J_{\theta}^{\top} J_{\theta}$ with $r = \text{rank}$ from (6) and trace bound (7),

$$\lambda_{\min}^+(J_{\theta}^{\top} J_{\theta}) \leq \frac{\text{tr}(J_{\theta}^{\top} J_{\theta})}{\text{rank}(J_{\theta}^{\top} J_{\theta})} \leq \frac{\tilde{L}^2 K \cdot N_{\text{data}} HW \cdot \text{tr}(\Sigma_X)}{N_{\text{data}} HW (N_{\text{cls}} - 1)} = \frac{\tilde{L}^2 K \cdot \text{tr}(\Sigma_X)}{N_{\text{cls}} - 1}.$$

Setting $C_2 := \tilde{L}^2 K \cdot \text{tr}(\Sigma_X)$ yields claim (ii) of the lemma. The constant C_2 depends on the input bottleneck dimension K , the classifier’s depth-dependent input Lipschitz constant $\tilde{L}$, and the trace of the data covariance $\text{tr}(\Sigma_X)$ —but not on C_g , the width M , or the spatial parameter count. This completes the proof. $\square$

Remark 33 (Corollary for per-pixel classifiers). When g_{ϕ} is per-pixel (no spatial mixing), the per-pixel classifier Jacobians $A_i \equiv A$ coincide, and the aggregated block factors as $J_{\theta}^{\top} J_{\theta} = A^{\top} A \otimes \Sigma_X$ (per example). Bhatia’s Kronecker spectral identity Bhatia [1997] then tightens the smallest positive eigenvalue to

$$\lambda_{\min}^+(J_{\theta}^{\top} J_{\theta}) = \mu_r(A^{\top} A) \cdot \sigma_{\min}^+(\Sigma_X),$$

where μ_r is the smallest nonzero eigenvalue of $A^{\top} A$. Both factors are C_g -independent. This tighter Corollary is what our synthetic MLP experiments verify (Experiment 1.1); the general Lemma 16 bound is what applies to real CNN/ViT spatial classifiers.

Remark 34 (Effective rank vs. rank ceiling). Lemma 16(i) is a bound on the rank ceiling, not on the effective rank. Neural collapse Papayan et al. [2020] predicts that in the terminal training phase only $\approx N_{\text{cls}} - 1$ of the $N_{\text{data}} HW (N_{\text{cls}} - 1)$ residual directions carry non-negligible curvature, so the effective rank is much smaller than the ceiling. Our bound is therefore loose as a descriptor of curvature-carrying directions but tight as the rank ceiling; either interpretation suffices for Theorem 11, since both imply $\lambda_{\min}^+(J_{\theta}^{\top} J_{\theta})$ is C_g -independent.

Remark 35 (Softmax vs. sigmoid outputs). The rank bound $HW(N_{\text{cls}} - 1)$ depends on the simplex constraint $\mathbf{1}^{\top} p = 1$, which holds for softmax and is preserved under label smoothing. For sigmoid / independent-Bernoulli multi-label losses the bound loosens to $HW N_{\text{cls}}$ (no simplex constraint). Our spectroscopic pipelines use softmax throughout, so the tighter bound applies.

A.1.4 Combining the three lemmas: full proof of Theorem 11

Assemble Lemmas 12, 14, and 16 to complete the proof of Theorem 11. Recall the joint Gauss-Newton matrix

$$G = \begin{pmatrix} J_{\theta}^{\top} J_{\theta} & J_{\theta}^{\top} J_{\phi} \\ J_{\phi}^{\top} J_{\theta} & J_{\phi}^{\top} J_{\phi} \end{pmatrix}$$

and the effective condition number $\kappa_{\text{eff}}(G) = \lambda_{\max}(G)/\lambda_{\min}^+(G)$ (Remark 10).

Upper bound on $\lambda_{\max}(G)$ direction, from the spatial block. Lemma 12 establishes

$$\mathbb{E}[\lambda_{\max}(J_{\phi}^{\top} J_{\phi})] = \Omega(M) = \Omega(\sqrt{C_g})$$

under Assumptions 3–4 for fixed-depth architectures with $C_g = \Theta(M^2)$, with concentration in the wide-network limit (Karakida et al. [2019], Theorem 5.1). Lemma 14 (Cauchy interlacing applied to the diagonal block $J_{\phi}^{\top} J_{\phi}$ of G) transfers this to the full matrix:

$$\lambda_{\max}(G) \geq \lambda_{\max}(J_{\phi}^{\top} J_{\phi}) = \Omega(\sqrt{C_g}).$$

Denote the implicit constant by c'_1 : $\lambda_{\max}(G) \geq c'_1 \cdot \sqrt{C_g}$ for all C_g sufficiently large.

Upper bound on $\lambda_{\min}^+(G)$ direction, from the spectral block. Lemma 16 (Step 6 of its proof) gives, under Assumption 1,

$$\lambda_{\min}^+(J_{\theta}^{\top} J_{\theta}) \leq L^2 \cdot \sigma_S(\Sigma_X),$$

where L is the Lipschitz constant of g_{ϕ} 's gradient (Assumption 3) and $\sigma_S(\Sigma_X) = \lambda_{\min}(\Sigma_X)$ is the smallest data covariance eigenvalue. By Cauchy interlacing (min form, Bhatia Bhatia [1997] Corollary III.1.5, dually):

$$\lambda_{\min}^+(G) \leq \lambda_{\min}^+(J_{\theta}^{\top} J_{\theta}) \leq L^2 \cdot \sigma_S(\Sigma_X).$$

Denote this upper bound by $C_2 := L^2 \cdot \sigma_S(\Sigma_X)$. Crucially, C_2 depends only on the model class of g_{ϕ} (via L) and on the data distribution (via σ_S)—it does NOT depend on C_g or on the number of spatial parameters in any way.

Combining the two bounds. Dividing,

$$\kappa_{\text{eff}}(G) = \frac{\lambda_{\max}(G)}{\lambda_{\min}^+(G)} \geq \frac{c'_1 \cdot \sqrt{C_g}}{C_2} = \frac{c'_1}{C_2} \cdot \sqrt{C_g}. \quad (8)$$

This is the essential result: the effective condition number of the joint Gauss-Newton matrix grows at least as $\sqrt{C_g}$ (equivalently, linearly in the spatial module's characteristic width M), with a rate constant c'_1/C_2 determined entirely by the classifier's regularity and the data statistics.

Recasting in the $\sqrt{C_g/C_f}$ form of Theorem 11. Assumption 1 sets $C_f = S \cdot K$, both fixed architectural constants of the spectroscopic pipeline. Multiplying and dividing (8) by $\sqrt{C_f}$:

$$\kappa_{\text{eff}}(G) \geq \frac{c'_1}{C_2} \cdot \sqrt{C_g} = \left(\frac{c'_1 \cdot \sqrt{C_f}}{C_2} \right) \cdot \sqrt{\frac{C_g}{C_f}} = c_1 \cdot \sqrt{\frac{C_g}{C_f}},$$

where $c_1 := c'_1 \cdot \sqrt{C_f}/C_2 = c'_1 \cdot \sqrt{SK}/(L^2 \sigma_S(\Sigma_X))$ is the theorem's leading constant, depending only on the data covariance (σ_S), the bottleneck dimension K , the input spectral dimension S , and the model class of g_{ϕ} (through L)—exactly the dependencies claimed in the theorem statement. The subtracted constant c_2 absorbs finite- C_g corrections from the mean-field concentration bound in Lemma 12. This completes the proof. $\square$

Remark 36 (Width vs. parameter-count scaling). The $\sqrt{C_g}$ scaling reflects the fact that Karakida's underlying result Karakida et al. [2019] bounds $\lambda_{\max}$ in terms of the network's characteristic *width* M , not its total parameter count. For a fixed-depth architecture with $C_g = \Theta(M^2)$, the equivalent statement is $\lambda_{\max} \sim \sqrt{C_g}$. The paper's Experiment 1.1, which sweeps width D from 16 to 8192 at fixed depth, measures a log-log slope of ≈ 0.70 against C_g —consistent with the asymptotic 0.5 ($\sqrt{C_g}$) prediction plus a mild finite-width upward correction of the same character reported in Karakida's own numerical validation.

Remark 37 (Why the $\sqrt{C_g/C_f}$ framing). Although $C_f = SK$ is a constant of the setup and the bound $\kappa_{\text{eff}}(G) \geq \Omega(\sqrt{C_g})$ is equivalent to $\Omega(\sqrt{C_g/C_f})$ up to constants, we prefer the ratio form because it makes explicit that the ill-conditioning is fundamentally a *capacity asymmetry* phenomenon: doubling both C_g and C_f (e.g. by widening both modules together) leaves the pathology unchanged. It is the ratio, not the absolute size, that drives the theorem.

A.2 Full proof of Theorem 26

[TODO: Phase 3, weeks 8-10. Combine Borkar singular perturbation with Pezeshki residual decay. Proof skeleton: (i) ODE limits of SGD, (ii) timescale gap from Lemma 5.1, (iii) Tikhonov fast-manifold convergence, (iv) NTK residual decay along fast manifold, (v) slow-variable bound by Gronwall.]

A.3 Empirical robustness

We here report robustness experiments referenced in the main text.

A.3.1 Optimizer ablation (SGD vs Adam)

[TODO: Phase 5 follow-up. Show EGR collapse pattern in both regimes; note Adam masks the natural gradient ratio.]

A.3.2 Loss-variant ablation

[TODO: Phase 5 follow-up. Cross-entropy, focal loss, label smoothing.]

A.3.3 Regularizer ablation

[TODO: Phase 5 follow-up. Weight decay, dropout, batch norm. Pezeshki et al. Pezeshki et al. [2021], App. B, already show gradient starvation is robust to these; we reproduce briefly in our setting.]

A.3.4 Smallest capacity ratio where the pathology disappears

[TODO: Phase 5 follow-up. Sweep C_g/C_f from 1 to 1000, identify the threshold below which joint training reaches frozen performance.]